Pei Wang Senior Software Engineer

Summary

- 15+ years of experience in Java, Python, .NET/C#, TypeScript, and SQL across the SDLC, building low-latency distributed systems, cloud applications, and data pipelines.
- Professional experience in real-time programmatic ad-tech, distributed systems, Kafka, cloud services (GCP/AWS), Kubernetes, data warehouses (Snowflake/BigQuery/Vertica), data pipelines (dbt/Airflow), SOAP/REST web services, SOA/ESB/EIP, and Loki/Prometheus/Grafana.
- Hands-on with Claude Code, Cursor, agent skills, MCP, and Gemini/OpenAI APIs.
- Collaborate easily with co-workers and work well independently. Proven resourcefulness to identify, analyze, and solve problems. Self-motivated and quick to learn new technologies.

Projects

Creator & Developer at ThinQr AI (AI homework tutor, Chrome Web Store) 2026

- Built and launched a Chrome extension (React/TypeScript, MV3) that tutors with escalating hints, not answers, reading problems off the screen via Gemini/OpenAI.
- Privacy-first, backend-free design: on-device memory, practice from past mistakes, bring-your-own API keys, on-demand permissions, and signed offline licenses.

Professional Experiences

Awards: DV Engineering Spot Award (Jan. 2025); 2021 Q3 DoubleVerify "All In" best employee

Senior Software Engineer II at DoubleVerify Jun. 2019 – Present

- Developed Ads programmatic pre-bid platforms that meet a 3 ms response time at the 99th percentile across 200+ billion ad auctions/day for 20+ DSP partners.
- Developed new architectures to improve the stability, scalability, availability, and memory efficiency of Ads pre-bid systems.
- Played a critical role in developing and maintaining high-concurrency, scalable, distributed systems across many tech stacks: Java, Kafka, NIO web servers, Chronicle Map, Aerospike, Ignite, Vertica, Snowflake, BigQuery, Spark, GCP, et cetera.
- Collaborated with PMs and built the first and best Ads Authentic Brand Safety (ABS), Contextual Targeting (CT), and Connected TV (CTV) pre-bid verification systems.
- Designed and implemented concurrent cloud applications and data pipelines on GCP/Kubernetes, including The Trade Desk and Amazon DSP reporting integrations (Python, Snowflake, dbt, Airflow, Terraform).
- Led AI-assisted engineering by building Claude Code/Cursor agent skills and MCP integrations that automate partner integrations, SQL migrations, and regression tests.
- Built monitoring and alerting with Prometheus, Grafana, and Loki, and drove CI/CD releases (GitLab, Flyway, Liquibase) across on-prem and GKE clusters.

Application Developer at MSIG Holding (Americas) Feb. 2017 – Jun. 2019

- Designed high-concurrency ESB/EIP architectures (Java) integrating legacy and new enterprise apps across UX/UI (Guidewire Claim Portal), middleware, and databases.
- Led design and development of CXF web services/APIs, ActiveMQ message queues, OSGi Karaf containers, and .NET workflows integrating Guidewire and Laserfiche systems.
- Developed .NET applications to automate insurance documentation.

Senior Developer at Direct Success Inc., Farmingdale, NJ Jan. 2016 – Feb. 2017

- Built .NET Windows apps and C# extensions for CPR+, a pharmacy management system.
- Developed C#/ASP.NET web apps, services, and APIs: Alert Services, Pharmacy HUB, Extra Step Assurance for medical cannabis dispensaries, and Vetlocity dashboards.

Senior Developer at Anichem Inc., North Brunswick, NJ Apr. 2010 – Jan. 2016

- Developed ISSolution (issolution.com), ASP.NET MVC web apps, and .NET desktop apps.

Internship at ENZO Biochem, Farmingdale, NY Jun. 2009 – Dec. 2009

- Developed drug-discovery bioinformatics and molecular modeling using SAS and SQL Server.

Personal Info

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Education

Ph.D.
Computational Organic Chemistry
State University of New York at Stony Brook, May 2010

B.S.
University of Science and Technology of China, Hefei, China, September 2003

Skills

Distributed Systems, Microservices
Service-Oriented Architecture (SOA)
Application Architecture, Web Services
Agile Methodologies
SOAP/REST, GraphQL
Java/Groovy/Scala
Python
C#/.NET/ASP.NET
TypeScript/JavaScript/Node.js
Spring Boot/JPA, Spring Framework
Apache Kafka, Pub/Sub, Ignite, Helios
Aerospike, Chronicle Map
SQL/NoSQL/Snowflake/BigQuery
Vertica, SQL Server, PostgreSQL
dbt, Airflow, Spark
Docker/Kubernetes (GKE)
AWS/GCP
Terraform, Flyway, Liquibase
GitLab CI/CD, Git, Gradle, Maven
Apache Camel, Apache Karaf, ActiveMQ
ESB/EIP, OSGi, XML/XSD/XSLT
React, Angular, HTML/CSS
Chrome Extensions (Manifest V3)
Splunk/Grafana/Prometheus/Loki
Claude Code, Cursor, MCP
Gemini/OpenAI APIs